

The exciting potential and daunting challenge of using GPS human-mobility data for epidemic modeling

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Francisco Barreras¹✉ & Duncan J. Watts^{1,2,3}✉

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Large-scale GPS location datasets hold immense potential for measuring human mobility and interpersonal contact, both of which are essential for data-driven epidemiology. However, despite their potential and widespread adoption during the COVID-19 pandemic, there are several challenges with these data that raise concerns regarding the validity and robustness of its applications. Here we outline two types of challenges—some related to accessing and processing these data, and some related to data quality—and propose several research directions to address them moving forward.

The field of human-mobility science has seen substantial advancements in recent years with the rapid adoption of Global Positioning System (GPS) location data, a trend accelerated during the coronavirus disease 2019 (COVID-19) pandemic. This shift has not only transformed our understanding of human movement but also revolutionized the modeling of epidemic spread. Although epidemic modeling has utilized human-mobility data on air and other transportation networks for decades^{1–3}, the scale and precision of time-stamped location traces from millions of mobile devices provide unprecedented representations of human-to-human interaction and mobility patterns. Unlike coarser mobility datasets, which are primarily useful during the initial outbreaks for the analysis of importation times and travel restrictions⁴, GPS location data can be used to analyze and monitor community transmission within cities and neighborhoods once an epidemic has reached pandemic status.

The ability to directly observe trajectories and contacts in real time also offers an appealing alternative to enormously complex agent-based models that attempt to anticipate changes in local mobility under various plausible interventions^{5–9}, especially in fast-moving policy environments in which, say, ‘stay at home’ orders can produce profound shifts in mobility patterns on the timescale of days to months. Finally, GPS location data are more scalable than Bluetooth solutions from wearable devices, and more precise than call detail records (CDR) or extended detail records (XDR)^{10,11}.

Reflecting their considerable potential, GPS mobility data have, in just the past three years, powered numerous exciting applications.

These developments were possible owing to several ‘data for good’ initiatives deployed in response to the COVID-19 pandemic¹². Applications include monitoring social distancing and mobility flows^{13,14}; identifying high-risk locations and behaviors for targeted interventions^{15,16}; exploring the susceptibility and compliance of different socio-demographic groups to infection and mandates^{17–22}; and assessing the effectiveness and optimal timing of non-pharmaceutical interventions^{4,23–25}.

Although it appears that this class of data will continue to be integrated into studies of epidemics and human mobility, its origins in the ‘location intelligence’ industry raise serious concerns regarding its collection and processing, and the resulting potential for biased and misleading inferences^{11,26,27}—not only in epidemiology but also in human-mobility research as a whole. Specifically, there are two classes of problem. The first class concerns the infrastructure limitations to accessing and processing the data, which in turn limits the transparency and validity of commonly used mobility metrics. As a consequence, researchers must choose between relying on non-transparently pre-processed metrics whose validity cannot be checked, and engaging in costly and time-consuming negotiations with the data providers to secure the underlying data and implement the processing themselves. The second class concerns data-quality issues—for example, its extreme sparsity and corresponding likelihood of bias—that result in errors in mobility metrics and downstream applications. Some techniques and good practices to address these quality issues, and to understand biases in passively gathered datasets, have been proposed in the past two decades—for instance, in the context of psychology

¹Department of Computer and Information Science, University of Pennsylvania, Philadelphia, PA, USA. ²Operations, Information and Decisions Department, Wharton School, University of Pennsylvania, Philadelphia, PA, USA. ³Annenberg School of Communication, University of Pennsylvania, Philadelphia, PA, USA. ✉e-mail: fbarrer@sas.upenn.edu; djwatts@seas.upenn.edu

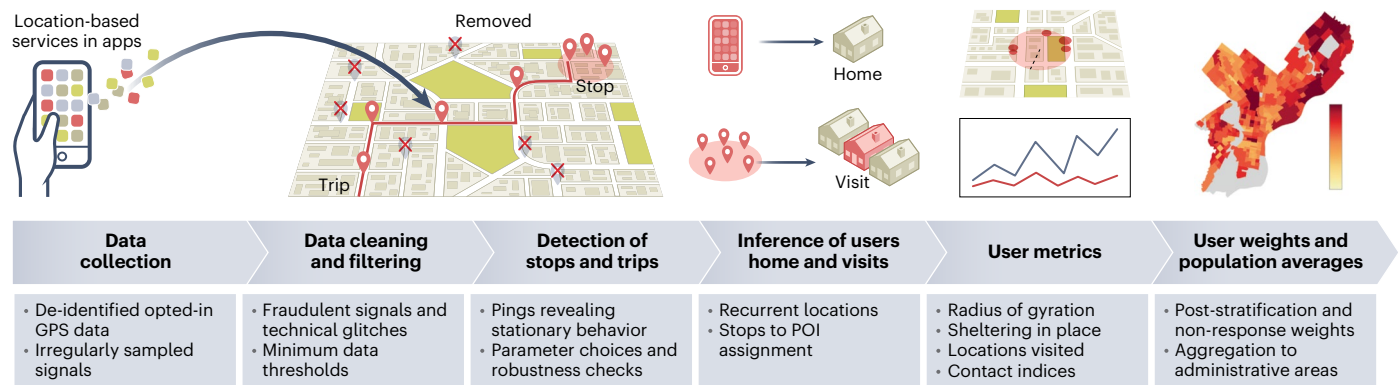


Fig. 1 | Broad overview of the data-processing pipeline. Example steps involved in estimating population-level mobility metrics, which include gathering data, cleaning, detecting stops and trips, inferring user attributes, computing metrics and, finally, averaging the results using user weights to obtain population-level estimates.

or recommender systems^{28–31}, or studying biases in CDR, XDR and geo-tagged social media datasets^{32–36}. However, researchers still lack generally agreed-upon standards for addressing data missingness and biases in massive GPS human-mobility datasets.

In this Perspective, we explore these two classes of challenges in more detail, along with their implications for research employing GPS location data in general and for epidemic modeling in particular. First, we discuss the infrastructural needs for data accessibility and good practices in the processing pipeline that leads to mobility metrics. Second, we discuss data-quality issues and the different types of error that are introduced in mobility metrics as a consequence. Third, we discuss three related issues—concerning differential cross-national rates of coverage, design of validation experiments and the estimation of potential contacts as a proxy for disease transmission—that we believe should be high priorities for future research. We conclude by making several recommendations regarding the development of standards and practices for accessing, processing and analyzing GPS mobility data that we believe will alleviate the challenges we identify. We hope that our discussion offers useful guidelines to researchers, policymakers and data providers on how to increase the transparency, quality and standardization of policy-relevant metrics and predictions, ultimately leading to more reliable tools for the management of future epidemics.

Challenges with accessing and processing the data

Boasting a market size of US\$16 billion as of 2022, the location-intelligence industry is a rapidly expanding sector comprising many individual data providers^{37,38} that we classify into two broad categories. In the first category are a handful of big players such as Google, Meta, Baidu, Foursquare and Apple, who primarily leverage data from their first-party suites of apps and services. In the second category is a much larger number of smaller companies such as SafeGraph, Cuebiq (now Spectus), X-mode and PlacelQ (now Precisely), who offer data that are gathered passively from potentially thousands of smartphone apps in which users have opted in to location sharing and services implemented using software development kits (SDKs). In some cases, these companies also buy data from intermediate companies that serve as ‘data aggregators’ instead of, or in addition to, gathering their own.

Regardless of how they obtain them, companies must somehow process the raw, device-level data to construct ‘mobility metrics’ that summarize individual- or population-level movement. Examples of such metrics include the radius of gyration^{22,39,40}, number of unique locations visited^{41–43}, indices of distance traveled^{44–46}, and mobility flows^{15,47–50}. Although these generic mobility metrics are relevant to

epidemic modeling, the COVID-19 pandemic also spurred the development of novel metrics that more explicitly capture interactions between individuals and hence serve as proxies for the risk of disease transmission. Some examples of these metrics are granular epidemic networks based on visits to places of interest (POIs)¹⁵, risks of transmission at the POI level^{16,51}, metrics of time spent at home or at other individuals’ homes^{52–55}, and estimates of potential human contact^{40,56–58}.

Importantly, most of the mobility metrics used in epidemiology research are pre-computed by the data providers themselves, sometimes in collaboration with research groups. SafeGraph’s Patterns dataset (<https://docs.safegraph.com/docs/weekly-patterns>), Descartes Labs’ Aggregated Mobility Index¹⁰⁸, Unacast’s Social Distancing Scoreboard (<https://www.unacast.com/post/the-unacast-social-distancing-scoreboard>), exposure indices using PlacelQ data⁵⁶, and Northeastern University Mobs Lab COVID-19 contact indices using Cuebiq data⁵⁹ are all examples of pre-computed, aggregated metrics. Metrics such as these are easy to access and hence were immediately useful to many researchers; however, they obscure a whole sequence of analytical choices that are typically proprietary and may also change over time in ways that are not transparent to the end user. Thus, independent verification of these metrics is impossible without access to the underlying data and methodologies. Using this data without properly assessing its representativeness, quality and biases could lead to invalid scientific findings or erroneous decisions by policymakers^{60–62}.

Here we show that the task of constructing and independently validating mobility metrics faces three infrastructural challenges: (1) a lack of standards and good practices with respect to analytical choices in processing the data, thereby undermining the validation and replicability of results; (2) barriers to access to granular datasets from a range of providers; and (3) a lack of access to standardized and validated methods for processing and analyzing this class of data.

Fragility with respect to analytical choices

The process of transforming raw GPS data into human-mobility metrics requires pre-processing and the application of algorithms in multiple stages. Raw data are typically composed of device-level sequences of time-stamped coordinates, or ‘pings’. The specific format of these records can vary but an illustrative example is ‘(id, lat, lon, t)’, where ‘id’ is a device-level pseudo-anonymous identifier, ‘lat’ and ‘lon’ are latitude and longitude coordinates, and ‘t’ could indicate the time of the ping in Unix format. Converting this type of data into meaningful metrics requires a complex processing pipeline and no consensus has emerged on the design of such pipelines. The result is a diverse array of individual approaches that is confusing for researchers who are mostly interested in constructing mobility metrics or proposing epidemic models.

To illustrate these complexities, Fig. 1 shows the key steps involved in a typical processing pipeline. First, data providers aggregate raw data from multiple location-enabled smartphone apps. The data are then cleaned—removing inaccurate signals, duplicates and false records resulting from glitches or deliberate ‘spoofing’^{63,64}. This cleaned data is then further filtered to a sample of users from a certain geography and with a minimum amount of data points. An important step in pre-processing is making sense of the trajectories by detecting stops—groups of pings indicating stationary behavior—and trips—pings indicating displacement between stops^{29,65}. Researchers can then use such pre-processed trajectories to infer visits to POIs or identify recurrent areas as users’ places of residence. Following the creation of these intermediate representations, a range of statistics is derived, capturing individual behavior and contact patterns, which form the basis for mobility metrics. These metrics are then typically aggregated and rescaled to obtain statistics at the level of cities or finer sub-geographies such as neighborhoods or administrative areas.

Figure 1 highlights a fundamental problem that arises when trying to validate or replicate existing results—namely, there are many defensible ways to perform each of the steps in the pipeline, where at every step different choices of methods could potentially yield different outputs. The result is a so-called garden of forking paths⁶⁶, where any one set of choices appears reasonable when viewed in isolation, but where very different conclusions may be obtained by making equally reasonable but different choices.

As an example, consider two separate attempts^{55,56} by different research teams to use seemingly interchangeable device-level datasets—from PlacelQ and Cuebiq—to estimate the frequency of contacts between individuals in a population, a key parameter in epidemiological modeling. In ref. 56, the authors constructed a ‘device exposure index’ that measures human contact within commercial venues. To construct this index, the authors first filtered their sample to US users with pings on at least 11 days out of any 14-day window during a baseline period preceding the pandemic to ensure consistent activity throughout the analysis period. They then utilized pre-computed stops by PlacelQ, each associated with visits to specific POIs and a proprietary ‘attribution score’, where visits falling below a PlacelQ-recommended threshold were removed. Home locations were determined weekly for devices having a minimum of three nighttime visits to the same residential venue, using interpolation rules to deal with missing or inconsistent weeks. The sample of users was further filtered to those for which a home could be identified with sufficient confidence. Finally, the authors computed the daily contacts of each device occurring in commercial POIs, and took the average at the census block group (CBG) level. When computing these averages, the authors imputed missing data from users not generating pings in any given day by assuming they were sheltering in place at home. In subsequent analyses, the authors derived user demographics from their home’s CBG.

In ref. 55, the authors also used raw GPS data to create a similar mobility metric, an indicator of ‘household mixing’ (H), which measured the fraction of users visiting residences other than their own on any given day; however, they followed very different data-processing steps. For example, the authors implemented their own stop-detection algorithm based on DBScan, a popular clustering algorithm⁶⁷, rather than using the data provider’s pre-computed stops. They then applied a different home-attribution rule to that used in ref. 56, defined as the most visited location in a 2-week period and recalculated every month. Their filtering criteria were also different: for example, they removed users with activity on less than 4 days or less than 10 stops in a month; they removed stops shorter than 15 minutes or at ‘not residential’ locations defined as closer than 30 m to any POI in the Ordnance Survey POI layer; and their averages considered only users that had pings that day.

These two examples illustrate three points that apply broadly to analyses that leverage GPS location data to draw inferences about human-mobility patterns. First, researchers must make numerous

decisions in the course of processing their data, often in the absence of clear theoretical guidance or empirical evidence. Second, different authors may take very different—and equally reasonable—paths even when trying to accomplish similar goals using similar data. Third, these differences could result in qualitatively different conclusions about human mobility. For example, ref. 56 concluded that there were no qualitative differences in mobility based on education or race, while ref. 55 suggested that there was low adherence to the household mixing bans after March 2020 and that increases in residential mixing in 2021 were evidence of ‘overcompensation’ and ‘lockdown fatigue’. If each team had used the other’s methodology, would they have reached the same conclusions? With no easy way of answering this question, what should we make of them?

Considering that almost every mobility metric offered by providers has a similarly complex—and in, many cases, unknown—underlying processing pipeline, the problem of interpreting qualitative findings is likely to be widespread. Recognizing this problem, some progress has been made in validating and standardizing methodological choices: by defining the steps and semantics in the processing of trajectory data^{29,65}; by providing open code for processing algorithms with default values for their parameters^{68,69}; and by validating the quality of some processing algorithms^{35,70,71}. However, given the evident policy relevance of at least some conclusions arising from GPS location data, increasing the commensurability of different studies remains an ongoing challenge. At a minimum, implementing some good practices found in recent applications based on GPS location data could greatly improve the robustness of results. For example, researchers should explicitly justify their filtering and parameter choices, arguing that they are adequate given the accuracy of the signals or that they are consistent with related work⁷². Similarly, applications can include robustness experiments, both for the most relevant parameters^{40,73,74}, or for filters that drastically reduce the sample⁷⁵. Finally, validating mobility statistics^{15,72,76} or directly replicating results on multiple datasets would improve robustness and consistency. As we shall discuss next, however, this last approach is currently impractical in most cases due to the difficulty of accessing the necessary data.

Barriers to data access

GPS location data can be used to infer highly sensitive information about individuals, such as their home address, workplace, places of worship and even their participation in public demonstrations⁷⁷. Moreover, the high predictability of human mobility means that it is not possible to protect user privacy with simple deidentification^{42,78}; indeed, something as simple as the top four locations visited provides a sufficiently unique signature to reidentify most users^{79–81}. In response to these serious privacy concerns, industry groups have proposed removing or distorting data associated with sensitive places⁸², but a simpler and more common approach is simply to restrict access to all raw data (although aggregated metrics still have privacy risks⁸¹). Complicating matters, many data providers also regard both the data and their internal data-processing pipelines as proprietary and therefore wish to protect both to safeguard their commercial interests. As a result, researchers desiring access to the raw data collected by location-intelligence companies must negotiate dedicated data-sharing and non-disclosure agreements. These agreements often involve licensing fees, which can exceed several tens of thousands of dollars annually even for academic licenses. Even when companies are willing to make their data available for free, negotiating the terms of each agreement is complex and time-consuming from the perspective of both the data providers, who wish to protect their interests, and the research institutions, who need to navigate legal and regulatory concerns while preserving the ability of researchers to publish their findings^{10,83,84}.

Given these logistical, financial and legal barriers, many researchers understandably resort to using the pre-computed, aggregated metrics that providers are willing to release publicly. Even in instances

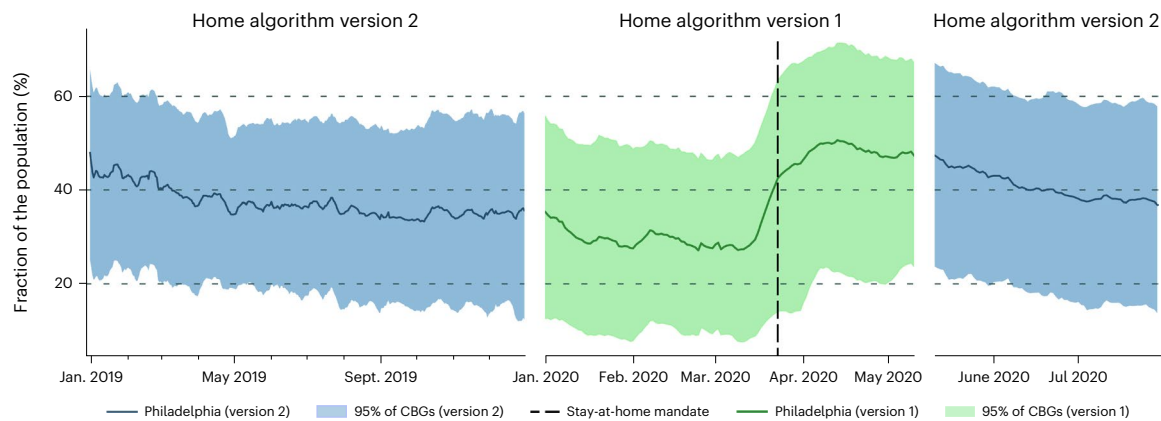


Fig. 2 | Fraction of the population in Philadelphia staying at home all day for 2019–2020 (7-day rolling average). The fraction of the population was estimated using SafeGraph’s social-distancing metrics (<https://docs.safegraph.com/docs/social-distancing-metrics>) and was computed at the CBG level, with the shaded areas containing 95% of the CBG-level values. An initial offering of this time series published daily in early 2020, using version 1 of SafeGraph’s home-

attribution algorithm, shows a reduction in mobility after the stay-at-home mandate was enacted. In May 2020, SafeGraph updated their home algorithm to version 2 for subsequent releases and offered 2019 data. Note that the values in early 2019 are substantially higher than those in January and February 2020, and are comparable to the values in June and July 2020. Also note that many CBGs have values exceeding 50% throughout 2019.

where researchers from well-funded institutions do negotiate a dedicated data-sharing agreement, it is unusual to go through this process with more than one provider; thus researchers are essentially never able to replicate their findings on more than one source of raw data. As reducing barriers to data access would likely enhance the reliability and robustness of research findings, we now briefly discuss some potential approaches for lowering these barriers.

Acknowledging the ethical concerns raised earlier, a promising strategy for improving data access is to enhance the privacy of the datasets shared with researchers and third parties, thereby reducing potential liabilities and relaxing the stringency of the agreed-upon handling of data. For example, one could check whether a specific analysis or model could proceed with coarser granularity. Providers could then apply spatial-temporal coarsening—also known as cloaking⁸⁵—before sharing the data. Notably, if an application proves to be robust to cloaking, CDR and XDR data—which have been more thoroughly studied for biases^{32–35,86}—could be preferred. While cloaking can hide users’ precise locations—reducing the sensitivity of the information exposed in a potential privacy attack—it, unfortunately, cannot guarantee that reidentification will be prevented^{87,88}. An alternative approach that does provide such guarantees, at least in theory, is differential privacy^{89,90}—a framework that adds random noise to queries such that no individual can be identified even when the querier has access to the rest of the dataset. Importantly, under certain conditions this noise can be added without biasing the data or substantially reducing its utility for inferring sufficiently aggregated statistics such as mobility metrics or networks for epidemic modeling^{91,92}.

Unfortunately, when highly granular or individual data are required—for instance, to study individual behavior, calibrate machine learning models, or simply because the aggregate metric that would solve a problem is not yet known—differential privacy is considerably less viable. The reason is related to the so-called curse of dimensionality, which arises when the regularities and correlations present in high-dimensional datasets would require adding too much noise to guarantee privacy, seriously degrading the utility of the data^{87,93}. A potential solution to this problem lies in leveraging deep learning techniques to generate synthetic datasets with joint distributions very similar to those of the sensitive data^{94,95}. The models generating these datasets can make use of differential-privacy techniques and can even be trained without centralizing the data using federated learning, thereby reducing the reliance on a trusted third party. We anticipate

that the combination of differential privacy and federated learning to improve the realism and privacy of existing synthetic trajectory generators^{96–98} will prove fertile ground for future research.

In addition to approaches for enhancing data privacy, a different set of solutions could come from new investments in data infrastructure augmented by standardized data agreements. For example, trusted research environments—secure, monitored and centralized platforms that allow researchers to conduct analyses directly on the provider’s servers—have been implemented successfully in clinical research using sensitive data^{99,100} as well as in epidemiological research¹⁰¹ and even by some location data providers¹⁰². These platforms can centralize the burdens of data governance, enforcement of aggregated and private outputs, and provision of computational infrastructure, thereby greatly simplifying data-sharing arrangements. The financial and logistical burdens, which disproportionately affect institutions with limited resources, could also diminish over time with market competition, especially if datasets prove interchangeable. Alternatively, transparent data collection by non-commercial entities—similar to contact-tracing, public-transit-ticketing or clinical-trial apps—could alleviate these costs^{31,103–106}. Perhaps the most practical solution might involve creating a comprehensive repository that contains mobility metrics, replicated across various datasets and geographies. Such a repository, unlike privately owned metrics, could have thorough documentation, underlying code and validation experiments, allowing the replicability of results without the need to repeatedly generate and validate inputs. A project of this magnitude would require coordination between data providers and researchers from multiple institutions and disciplines—human mobility, epidemiology, differential privacy—potentially building on existing initiatives (for example, covid19.flowminder.org and www.crisisready.io). The support of multilateral organizations such as the World Bank’s Data for Development program, which sees data as a ‘global public good’¹⁰⁷, could also help increase accessibility to these resources globally and equitably.

Lack of available methods for data processing

Even with a clearly defined methodology and access to a quality GPS human-mobility dataset, researchers still face daunting technical challenges when processing this type of data. Large-scale spatial data can amount to dozens of terabytes in volume¹⁰⁸. Applying the processing algorithms depicted in Fig. 1 to data of such scale often requires distributed computing solutions, such as high-performance computing clusters, big-data processing libraries such as Hadoop, GeoSpark

(Sedona) or Dask, and cloud-based services offering scalable and flexible data handling and storage capabilities. In this context, the essential requirements for researchers include not only access to advanced computing infrastructure but also algorithms that are compatible with these systems and designed for high efficiency and scalability. These requirements alone make these data inaccessible for researchers lacking expertise in big-data processing, especially when referencing publications that do not make their code publicly available (some exceptions include refs. 55,71,73,109). In the face of these challenges, open-source software libraries implementing the various transformations and analyses in the processing pipeline (Fig. 1) are essential to facilitate research for first-time users, as well as to enhance the replicability and robustness of existing methodologies. For example, Python libraries such as scikit-mobility⁶⁸, ArcGis Pro¹¹⁰ and trackintel⁶⁹ are robust and allow pre-processing, stop and trip detection, mobility metrics, and visualization. There are also libraries in R and C++ such as Tracktable¹¹¹, adehabitatLT¹¹² and TrajDataMining¹¹³ (see ref. 69 for a more comprehensive list) that implement similar functionalities, although these tend to be less complete than the Python versions. In addition, as raw CDR and XDR data are pre-processed in similar ways to raw GPS data, libraries such as Bandicoot¹¹⁴ and mobilkit¹¹⁵ could be easily adapted to work with the latter.

Moving forward, these libraries could implement existing alternatives for each of the processing algorithms involved so that different results in the literature can be replicated. Crucially, these libraries could extend their functionalities to allow for distributed computing. As many analyses can be vulnerable to high sparsity and noise, libraries could also implement methodologies that are suitable for this type of data—for example, building on the implemented sparsity metrics and filters in scikit-mobility⁶⁸ and trackintel⁶⁹, and developing tools for debiasing mobility metrics such as weights for stops and users that facilitate post-stratification procedures.

Challenges with data quality

When researchers rely on data that is pre-processed by data providers, the methods are rarely disclosed, and—being proprietary—might even be unknown to the researchers themselves. Biases resulting from the original data collection^{26,27} as well as those introduced in the data processing^{35,70,116} are therefore obscured, leading to puzzling anomalies and inconsistencies that are difficult or impossible to resolve.

For example, consider one of the widely used social-distancing metrics from SafeGraph, the daily number of devices staying completely at home^{15,52–54,117–119}, shown in Fig. 2, which depicts the percentage of the Philadelphia population staying at home all day. The initial release of these data, represented in green in Fig. 2, spans the first half of 2020 and appears to show that the city-wide stay-at-home mandate resulted in a dramatic increase in the fraction of people staying at home all day—a result that was cited in several studies early in the pandemic^{53,120–122}. However, subsequent releases of the data, which included historical data for 2019 (Fig. 2, blue regions) reveal two important anomalies of the time series that raise doubts about the standard interpretation. First, looking at the blue region on the left, which runs from January to December 2019, it appears that many parts of Philadelphia had experienced as many as 50% of residents remaining homebound—a number that seems implausibly high for the pre-pandemic era. Second, over the course of 2019, there is an unexplained downward trend followed by an somewhat sharper dip in early 2020. Overall, the drop in percentage staying at home that appears to have occurred between January 2019 and March 2020 in the absence of any public concerns about a lethal pandemic is about the same size as the subsequent jump arising from the lockdown. SafeGraph has attributed anomalies in their data to a change in their home-attribution methodology that was implemented in May 2020 and was applied to the 2019 release; however, this explanation does not elucidate either the spuriously high average rate of staying at home in January 2019 (which is roughly

the same as in June 2020) or the presence throughout 2019 of CBGs in which a majority of residents were apparently at home the entire day. Moreover, because the update was not backfilled to the key period of early 2020, the explanation is hard to evaluate.

Regardless of whether anomalies such as this one are explained by biases in the data or methodology changes, they raise serious concerns about the validity of the myriad epidemiological findings using this time series. And, although researchers have called for better validation of GPS and other digital-trace mobility data^{36,108,123–128}, external validation of the GPS-based metrics widely used in research remains virtually non-existent, hence the nature and severity of the data-quality problems are unknown.

As summarized in Fig. 3, we suggest that there exist at least three distinct but related issues that deserve much greater attention than they have received. First, in contrast to the marketing materials for most providers, which tend to emphasize the massive scale of their samples, GPS location data are extremely sparse, meaning that the vast majority of devices in any given population are absent (that is, fail to register any pings) most of the time. Second, the missing data are almost certainly not missing at random. On the contrary, as we will discuss, whether or not a device pings (and hence appears in the sample) depends very much on a variety of factors such as access to Wi-Fi, app activity and even movement itself that are directly related to the outcome of interest. Such ‘sampling on the dependent variable’ therefore creates potentially severe bias in the observed data, greatly complicating efforts to impute missing data. Third, additional bias can be introduced when applying processing algorithms that are not robust to parameter changes or to the sparsity inherent to this type of data. These three sources of error—sparsity, bias, and lack of robustness in processing algorithms—as well as how they manifest empirically in the downstream analyses and mobility metrics, are the next focus of our discussion.

Sparsity

The first of these issues, data sparsity, is rarely addressed in the literature^{40,56}. Most work simply references the number of devices included in the sample without specifying the inclusion criteria or testing the robustness of results to different filters. Mobility metrics for users with very few pings can be unreliable and—at least for some metrics such as the radius of gyration, the number of unique locations visited, or the mobility entropy^{42,129}—might require adjustments to remove bias. However, it is often the case that these ‘highly incomplete’ users are counted in the coverage metrics reported by data providers. This high sparsity is evident when comparing the monthly active users (MAU) and daily active users (DAU) in samples of data offered by human-mobility-data providers. The ratio of DAU to MAU—shown in Fig. 4 for a few samples for which those statistics are publicly available—provides an upper bound for the percentage of activity measured for a typical user. In practice, however, the device-level coverage can be as low as only a handful of hours in a whole month.

We note that this extreme missingness of data does not necessarily pose a serious problem for the data providers themselves, who are generally in the business of serving location-aware ads to specific devices (for instance, a coupon for McDonald’s when the device pings near a specific McDonald’s franchise). As long as they can correctly infer that a device is near a particular location at a particular time, it does not matter that they do not know where it is at any other time. For the purpose of accurately estimating epidemic spread, however, it is critical to account for all devices at all times, or at least to have an estimate of the bias in the sample of device locations such that missing data can be imputed or metrics can be adjusted. We will discuss below experiments that can show that these samples are biased; however, it is important to understand that even if the data were sampled without bias, the sparsity alone introduces errors and induces trade-offs that complicate the processing pipeline. These complications arise from

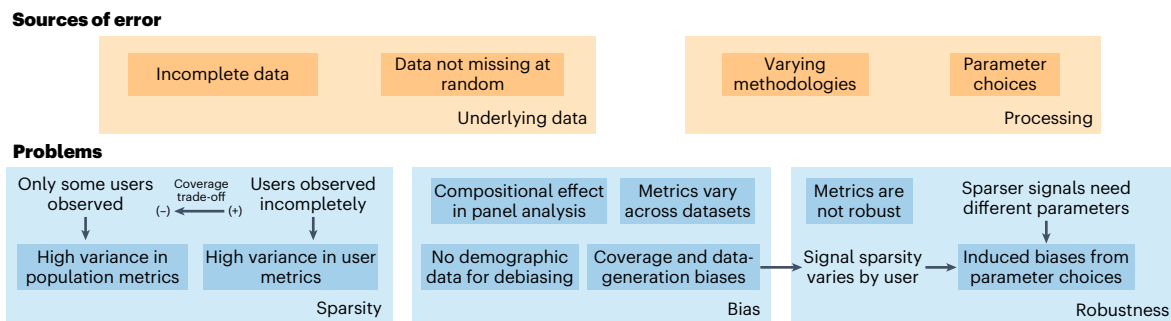


Fig. 3 | Sources of error in the estimation of metrics from GPS location

data. Errors in estimating metrics may arise at various stages of the processing pipeline. They can result from the underlying data being sparse and biased, from the selection of methodologies that may amplify or induce biases, or from the

choice of parameters within those methodologies that prove inadequate for handling sparse trajectories. These errors can be studied by using statistics of the data sparsity, testing for biases and conducting robustness tests.

the fact that the sampling error in population mobility metrics—the variance coming from having only a finite sample—is inversely proportional to the sample size. On the one hand, this fact implies that larger samples of individuals will yield more robust estimates; however, it also holds at the device level, wherein users with more incomplete data will have higher error in their individual estimates. Hence, regardless of sampling bias, the high sparsity means that decisions to filter out users with few data points come at a cost and require further scrutiny.

Another concerning feature of the data is that the number of active users can change dramatically over time depending on behavioral variables—for example, staying at home, traveling or particular app usage—or choices by data aggregators (for instance, acquiring or dropping partner apps). These changes in panel size are often acknowledged by researchers—for example, an attrition rate of over 85% of the sample in ref. 55—but are challenging to deal with. The reason is that all of these sources of variation can lead to ‘compositional differences,’ meaning that apparent changes in population-level estimates are driven by changes in the composition of the sample rather than any changes in behavior¹³⁰. Although it is possible to safeguard against the resulting biases—say, by restricting the sample to users that appear consistently—these methods again encounter a trade-off between bias and coverage^{56,57}. Future studies should address this trade-off, potentially using techniques such as weighting or imputation to make incomplete trajectories usable and comparable to complete ones^{31,131–133}. At a minimum, data providers and researchers alike should be more transparent about the completeness of human-mobility datasets, potentially using a variety of metrics that assess coverage at different intervals—for example, hourly, daily or weekly. Finally, population-level averages should also be corrected to account for missing data. While techniques such as post-stratification or raking have been used in the human-mobility literature^{15,17} to address geographical biases, adjustments due to missing users are rarely implemented⁵⁶. The effect of user weights on epidemiological metrics such as contact indices can be critical, as the contact of two individuals should count as the product of their weights. In such cases, trimming is advised to prevent unstable estimates from extreme weights¹³³.

Biases

Sparsity is not the only source of bias with this class of data. Previous work has already highlighted several biases that are inherent to collecting GPS human-mobility data^{11,26,27}. Some of these biases are related to the demographic and geographic distributions of smartphone ownership. For instance, the data tend to over-represent younger, higher-income, higher-education and urban populations^{33,134}, while under-representing geographic areas with low smartphone penetration, limited internet access and where the costs of data plans can be prohibitive—particularly, in developing nations where there is

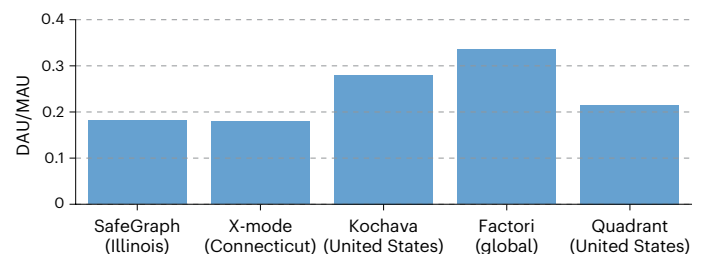


Fig. 4 | Ratio of DAU to MAU for five samples of GPS human-mobility data.

These ratios, ranging between 0.18 and 0.33, indicate that the majority of users have no activity during a majority of days and provide upper bounds for the typical coverage at the device level. The data sources for this figure are detailed in the ‘Data availability’ section.

also substantial heterogeneity in phone ownership and usage^{135–137}. Beyond phone ownership, it is challenging to identify other attributes in samples of users of GPS location datasets—which lack any personal identifying data—that can be used to detect biases. Some researchers attempt to detect these biases by imputing demographic attributes from the census areas where devices are predicted to reside^{40,56,117,138–143}. Studies of this type have found potential biases that vary across data providers (for example, ref. 56 found that PlaceIQ data over-represent areas of low population density and large shares of white residents in the United States, while ref. 144 found the opposite biases in SafeGraph data). However, these analyses might be inadequate given the well-known inaccuracies in home-attribution algorithms^{35,70}, and the strong assumption of homogeneity within census areas when imputing demographics¹⁴⁵. A way to avoid this ‘ecological fallacy’ is to infer demographics from much more granular data, such as foot-traffic to voting locations linked to voter-roll data¹⁴⁵.

While the aforementioned biases relate to which users are in the sample, other biases relate to how much of their data is measured—which is affected by technical and behavioral factors alike. The luxury nature of smartphone ownership and data plans in parts of the developing world also implies that users (and regions) with lower income will generate less data and in less passive ways^{135,136}. Even for users without these limitations, the power-hungry nature of GPS tracking¹⁴⁶ can lead operating system software to halt or to reduce the rate of collection of GPS measurements when connected to a home Wi-Fi network, when there is low battery or when no ‘significant changes in location’ are detected^{147,148}—all of which are probably correlated with human mobility. Biases can also arise from behavioral factors such as a user’s propensity to opt-in to data sharing, which in turn is associated with traits such as agreeability, risk-taking propensity, age and education^{149–152}, as well as the fact that data might be collected from specific apps associated

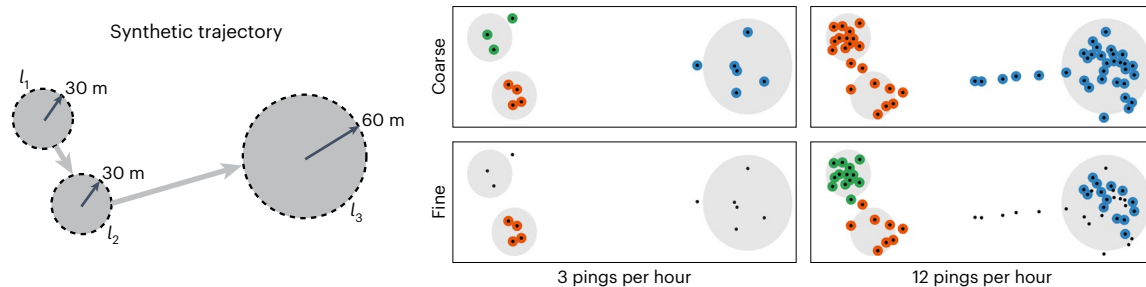


Fig. 5 | Effect of stop-detection algorithm parameters on a synthetic trajectory. Two parameterizations of DBScan are applied on GPS traces sampled at different frequencies from the same synthetic trajectory, consisting of consecutive visits to locations l_1 , l_2 and l_3 . Within each panel, different colors represent detected clusters. For a specific ping frequency, there is a

range of parameters that is most appropriate for extracting the true stays. In our simulations, we used the following parameter combinations: the coarse parameters are $\epsilon = 120$ m and $\text{min_pts} = 2$, and the fine parameters are $\epsilon = 75$ m and $\text{min_pts} = 3$.

with mobility or other behavior—for example, transportation versus entertainment apps¹⁰⁸. Ideally, debiasing would involve knowledge of the publishing apps and sourcing from diverse apps for each individual. Unfortunately, such knowledge is almost impossible to extract from the current data-gathering pipeline (see ref. 153 for an effort to detect the trackers in some of these apps), which has been described as “long and opaque”²⁶, so that “each actor [...] is able to buy, clean and transform data without providing enough transparency”.

Given the multitude of potential biases and the general absence of transparency in the data-generating process, detecting which biases are present and quantifying how serious they are presents a tremendous challenge for future research. For example, future work should: replicate and contrast results with landmark validation experiments on CDR and XDR data, which include comparing results with ground-truth data^{32,35,136}; evaluate the impact of debiasing methods on mobility metrics³³; and examine coverage and behavioral differences in populations without phones or using legacy phones. The latter is especially pertinent in the developing world, where smartphones have much lower penetration^{34,154}, potentially making CDR data more representative in these contexts. Together, studies such as these could reveal which biases affect analyses the most, which metrics are more robust, and which datasets are more representative.

Robustness of processing algorithms

As discussed previously, mobility metrics and results in the literature might lack robustness due to the multiple paths that can be chosen to process the data from raw GPS traces to results. Although non-robustness can arise in many ways, here we focus on choices regarding an algorithm’s parameters such as: the roaming distance and time-gap thresholds in stop detection^{29,116,155}; the time window and minimum stops required for determining home locations^{35,56,70}; density parameters for detecting locations from stops⁷¹; rules for attributing stops to POIs^{72,156}; speed parameters for cleaning or detecting travel modes^{68,69}; and proximity parameters to define co-location of individuals^{40,57}. Varying these parameters can lead to errors such as omitting or merging stops, mislabeling trips as stops, mistaking hotel stays as residences, misattributions of stops to POIs, and overestimation of transmission risk in epidemiological applications. It is rare that these parameter choices are disclosed in the literature (and even rarer to find robustness experiments) in large part because commercial providers tend not to disclose the proprietary methodologies underlying their pre-computed metrics. As a consequence, it is difficult to quantify errors of this type on mobility metrics, underscoring the need for further research to determine which algorithms and parameters are most appropriate.

Naturally, research of this sort will take time and—due to the great variety in data sources and the prospect of new algorithms that will emerge in the field—will probably not produce universally good

choices. Fortunately, increased transparency and replicability is possible even in the short run by adopting relatively simple practices such as disclosing and justifying the parameters used in processing. For example, parameter choices could be justified by replicating related work, by incorporating contextual information on the space where individuals are moving, or from analysis of the accuracy and sparsity of the signals. Alternatively, one could test algorithms using synthetic trajectories that incorporate realistic features of human mobility and sparse measurements similar to those in real datasets. Although simulated data do not constitute ground truth for these mobility metrics and algorithms, these experiments can be an intermediate step between heuristics and more rigorous validation literature. In what follows, we demonstrate an example of such tests for the task of stop detection, showing that sparsity and geographical context can decisively affect the output and could be used to calibrate parameters.

Although a variety of stop-detection algorithms exist¹⁵⁵, our example is based on DBScan⁶⁷ owing to its simplicity and use by both data providers and academic researchers^{55,72}. DBScan is parameterized by ϵ , which is defined as the maximum distance allowable between two points to be considered neighbors, and min_pts , defined as the minimum number of neighbors that a point must have to not be considered noise. Specifically, we study two sets of parameters that affect the resolution of the detected stops in a synthetic trajectory, namely ‘coarse’ parameters ($\epsilon = 120$ m, $\text{min_pts} = 2$) and ‘fine’ parameters ($\epsilon = 75$ m, $\text{min_pts} = 3$). The synthetic trajectory simulates two hour-long visits to adjacent 30-m-radius locations, l_1 and l_2 , followed by a trip to a 60-m-radius location, l_3 , where a two-hour visit takes place (Fig. 5). We sample signals at ‘low’ (3 pings per hour) and ‘high’ (12 pings per hour) frequencies to demonstrate the impact of having heterogeneous sparsity patterns in the data.

As illustrated in Fig. 5, the sparsity and parameter choices can affect the algorithm’s ability to detect stops and exclude trips. A coarser parameterization may struggle to differentiate stops at small neighboring establishments. Conversely, a finer parameterization might completely miss a stop at a larger location, mistaking it for short trips or noise. Looking at the effect of frequency, low-frequency trajectories seem less prone to stop merging, and could use coarser parameterizations to better cluster sparse pings, whereas high-frequency ones benefit from finer parameterization to avoid merging stops with trip pings.

Many of the issues observed in this toy example can probably be addressed with adequate pre- or post-processing. For instance, detecting trips before stops could prevent merging trip pings; coarser and finer clustering algorithms could be applied sequentially, splitting ‘big’ stays when appropriate; or a hierarchical clustering algorithm could be applied to deal with varying densities^{71,157}. Furthermore, the simulation underscores the importance of contextual information in parameter choices, perhaps treating dense urban areas differently, or finding parameters that equalize the outputs of high- and low-frequency users.

Many algorithms have been designed to try to minimize these problems in stop detection, and to eliminate the need to choose appropriate parameters^{116,158–161}. However, we stress that it is important that the suitability of these algorithms be demonstrated with validation experiments, and not assumed.

Temporal parameters in stop-detection algorithms face similar challenges and can also be difficult to set. A very small gap tolerance will result in splitting longer stops into multiple shorter ones, which can be undesirable if the user was indeed in a location continuously. However, bridging gaps that are too long—particularly for recurrent locations such as homes or workplaces—can result in errors due to unobserved short trips to other locations potentially happening during the gap. These changes in dwell times could have a considerable impact on metrics relevant for epidemiology—for example, by overestimating the time spent at home or underestimating the radii of gyration for users with sparse signals depending on the parameter choices. The issue might be amplified by the possibility that devices ping less frequently when they are idle, particularly at home⁵⁶; thus, it is good practice to consider the sparsity patterns of signals near recurrent locations when making these parameter choices.

Related concerns

To complement our discussion of data-access and quality challenges, we now explore three related issues that we believe should be central to the discussion on how to correctly integrate GPS location data into the modeling of epidemics: (1) addressing the unequal impact of data sparsity on policy and research, (2) designing better data validation experiments, and (3) estimating potential contacts for infectious disease transmission.

Addressing the unequal impact of data coverage gaps

The study of human mobility during the COVID-19 pandemic revealed important socioeconomic disparities in terms of economic impact^{162,163}, risk of transmission^{15,17} and the ability to practice social distancing^{20,164,165}. These disparities underscore another dimension for concern—the inequitable access to technological and infrastructural solutions for addressing future epidemics. Specifically, the bias and coverage issues we have highlighted—stemming from precarious access to smartphones, internet or data plans in certain regions of the world^{34,166}—can lead to increased errors in metrics and less reliable models and findings for regions of already-high vulnerability. Researchers wishing to work with higher-quality datasets might, inadvertently, reinforce these disparities by over-focusing on the more reliable datasets from developed countries.

However, the features of GPS location data also offer a unique opportunity to make human-mobility and epidemiological research more equitable through increased accessibility. Unlike CDR data, which are tied to local telecom providers, or government contact-tracing apps restricted by national boundaries, data from smartphone apps with presences in multiple countries can be aggregated by a single provider, which might even offer global datasets. We remark that allowing analyses of devices crossing national borders poses ethical concerns and, as a result, is not currently permitted by many companies. Even if cross-national aggregation is not possible, however, simply having access to data from multiple countries without requiring local connections should facilitate cross-national comparative research and improve coverage of regions that are historically marginalized in research applications. Naturally, these benefits are bound by the gradual penetration of smartphone devices and internet access, and the expansion of location-intelligence companies. Therefore, CDR datasets with higher coverage in data-poor regions will continue to play an important role.

Designing better data validation experiments

Given the potential policy implications of epidemiological models based on GPS location data, the ability to validate these datasets is a

clear priority for future research. As we have discussed, for example, de-identified human-mobility data can be merged with census demographics—through the use of home-attribution algorithms—to identify biases. Future studies could build on this work by expanding to larger geographical and temporal scales^{56,138,144} and performing more comparative studies across datasets. Furthermore, as behaviors such as air travel, attendance at events, or use of public transit can also be inferred from GPS traces, there is an opportunity to validate these data by joining with other datasets and surveys that measure those behaviors along demographic lines^{145,167–170}. Future surveys, capturing demographics and ground-truth trajectories from a representative sample, could be created for this purpose. While funding for this type of infrastructure investment might pose a challenge, such efforts would directly measure, rather than impute, demographics, enhancing validation robustness.

Another validation approach involves comparing mobility metrics across different datasets^{15,40,54,76,164}, as robustness with respect to the choice of dataset would increase confidence in these metrics. Future analyses should include periods of typical mobility and be more spatially granular as biases can be created or obscured by aggregation¹⁷¹ or by drastic shocks to mobility¹⁶⁴. It is also critical to use the same metrics and methods to ensure that differences stem from the datasets themselves. Although there is a tendency to focus validation works on urban areas due to the increasing concentration of population in cities, it is important to note that rural areas present unique challenges—such as lower smartphone and internet access, increased data sparsity, and, crucially, less data to compare against. Therefore, validation in urban settings should not be automatically assumed to extrapolate to rural areas, where data utility and biases need separate assessment.

Estimating potential contacts for infectious disease transmission

A key goal in epidemic modeling—particularly in the study of airborne diseases such as COVID-19—is to estimate when and where disease transmission events occur. GPS location datasets, due to their high precision, can achieve this goal more accurately than CDR and XDR datasets by better capturing co-location events. Consequently, recent literature has offered a variety of contact indices that can be more relevant for modeling epidemic spread than traditional mobility metrics based on movement and distances.

Importantly, defining these contact indices requires choices about the spatiotemporal precision; in particular, how close should two devices be in space and time to count as a co-location event? Also importantly, these definitions vary in the literature. For example, some work defines spatial proximity by tracing a radius around each GPS trace and finding intersections between devices—with radii ranging from 2 m (ref. 57) to 50 m (refs. 40,72). In contrast, other work builds contact indices using gridding, defined as counting how many devices fall in the same ‘cell’ of some chosen spatial subdivision such as geo-hashes⁵⁹ or Bing tiles¹⁷², while yet other work defines co-location as when two devices stop at the same POI^{15,16,173}. Similarly, researchers must also choose how proximate a co-location needs to be in time—for example, within the same 5-minute window⁵⁹, 1-hour window^{15,40} or a day¹⁷⁴.

Determining the appropriate precision is not trivial and, as with other choices we have discussed, faces a trade-off. On the one hand, epidemiological studies often focus on high-risk transmission events—such as those occurring within a 2 m range as per the Centers for Disease Control and Prevention guidelines—and on identifying POIs that are infection hotspots^{15,16}. On the other hand, achieving such high precision is fraught with challenges, some of which are technical—such as GPS signal noise⁵⁷, differentiating indoor versus outdoor signals, and the complexity of assigning stops to POIs in dense urban areas¹⁵⁶—while others are statistical, related to how the incompleteness of the data makes co-location a rare event. For example, if epidemic networks are constructed from highly precise co-location events then the resulting

BOX 1

Key needs to increase the utility of GPS human-mobility data

Infrastructure resources

Repositories

- An independently validated and well-documented compendium of human-mobility metrics from multiple datasets, increasing replicability and accessibility.
- Improved community-maintained repositories that centralize processing and analysis methods (such as refs. 68,69,115) that can handle noise and sparsity, scale to massive datasets and allow for different methodologies for robustness testing.

New datasets and data-access solutions

- Reduced barriers to access the underlying granular data, either technically, through solutions such as trusted research environments for ‘in-house’ analyses, or contractually, by standardizing data agreements at affordable academic pricing.
- New datasets capturing GPS traces gathered by non-commercial entities, increasing transparency.
- Synthetic trajectory datasets that replicate features of human-mobility and sparse data collection, allowing for testing of algorithms and code without incurring licensing costs.
- Increased access to surveys and administrative datasets—including future surveys with ground-truth trajectories—to be joined with de-identified GPS location datasets for validation.

Research

Standardization of methodologies

- Comprehensive studies describing and comparing alternative processing methodologies—such as estimation of stops, homes and flows—that will establish standards and good practices when processing highly sparse data.
- Methodologies to address the extreme sparsity in this class of data—applied especially in data-poor regions affected the most by it—measuring its effect in the robustness of metrics and algorithms.

Validation of data and algorithms

- Studies and applications leveraging the increased access to data from marginalized regions, to make epidemic-related advances more equitable.
- Testing for demographic bias by using granular datasets, other than census data, for better inference of user demographics and to avoid obscuring biases by spatial aggregation.
- Validating commercial POI datasets, with rich metadata, by comparing with potentially more complete data such as crowd-sourced datasets, census data and POIs detected from GPS traces.
- Comprehensive and transparent studies replicating the same methodologies for policy-relevant mobility metrics and findings across multiple datasets.

transmission rates between subpopulations would have increased sparsity and variance. Thus, researchers might opt to sacrifice precision to reduce said sparsity, without compromising the utility of the metrics. Using longer time windows could also aid in modeling the persistence of pathogens in the air, as well as fomite transmission¹⁷⁵.

Another important factor to consider when measuring co-location events is whether the methodology itself can preclude some of these events from being counted. The sample deficiencies already discussed

imply that contacts occurring in schools, prisons or nursing homes—all of which are high-risk hotspots of epidemic spread—will be substantially underestimated. Correspondingly, methodologies based on grids will grapple with the ‘boundary problem’ (where close signals occurring in different adjacent cells are not counted), while methodologies based on stop locations will ignore contacts occurring outside—excluding crucial vectors of disease transmission such as public transportation systems. Finally, if contacts are estimated using commercial POI datasets—like those from SafeGraph or Foursquare—many interactions might be excluded. Although these datasets offer rich and valuable POI metadata such as business categories, areas and operational hours that can be useful for estimating disease transmission^{15,16,56,72}, they may disproportionately represent certain types of venue, such as bars and restaurants, while omitting non-commercial places such as residences and workplaces. How much influence any given provider’s sample of POIs has on policy-relevant findings like the purported high-transmission risk of restaurants and gyms^{15,16} therefore remains an important unanswered question.

Facing these challenges, a promising research direction is to compare and validate these methodologies, verifying their consistency with existing literature and their correlation with epidemiological data and contact diaries^{51,57}. Some of the difficulties mentioned above could be addressed by following practices already present in the literature. For instance, boundary problems in gridding methods could be dealt with by employing adjusted counts from overlapping cells. Unobserved transmission events at schools or workplaces can be imputed with simulated data¹⁴¹. Finally, when using commercial POI datasets to measure human interactions, researchers are advised to not ignore potential contacts occurring outside of them. For example, by complementing their commercial datasets, which may have rich metadata, with POIs generated directly from GPS traces in an unsupervised way^{71,73}, or with crowd-sourced datasets⁷², which can be more complete and less biased. Similarly, transmission events in public transit could be estimated from smartphone ticketing apps¹⁰⁵, or estimated from GPS traces with trip segmentation tools¹¹⁵.

Recommendations and future work

Box 1 summarizes the main high-level recommendations for future research that we believe follow from our discussion. We note that while following these recommendations may require investments that are much larger than are typical in epidemiology and human-mobility science, the recent—and to some extent ongoing—COVID-19 pandemic should remind us that the stakes are even larger. In addition to the human and economic distress directly attributable to the pandemic itself¹⁷⁶, which was both enormous and global, COVID-19 exposed our collective vulnerability to epidemics of infectious diseases, revealing critical flaws in our response mechanisms and current epidemic modeling technology. Thus, addressing these challenges is not merely an academic exercise but an imperative for the efficacy of our responses to future public health crises. We hope our discussion can serve as a framework for a community-level dialogue on how to address these challenges moving forward.

Data availability

Mobile phone mobility data from SafeGraph depicted in Fig. 2 were made available to the research community through SafeGraph’s COVID-19 Data Consortium (<https://www.safegraph.com/covid-19-data-consortium>). The data for Fig. 4 can be accessed from the following sources: SafeGraph, <https://www.eff.org/deeplinks/2021/08/illinois-bought-invasive-phone-location-data-banned-broker-safegraph/X-mode>⁵⁷; Kochava, <https://www.ftc.gov/news-events/news/press-releases/2022/08/ftc-sues-kochava-selling-data-tracks-people-reproductive-health-clinics-places-worship-other>; Factori, <https://datarade.ai/data-providers/lifesight/profile>; Quadrant, <https://www.quadrant.io/solutions/data-quality-dashboard>. We have archived the websites referenced in this paper for future reference (Supplementary Table 1).

Code availability

Code for the evaluation of stop-detection on synthetic trajectories as shown in Fig. 5, as well as the visualization in Fig. 2, is available at <https://github.com/Watts-Lab/stop-detection-validation>.

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Author contributions

F.B. and D.J.W. conceived the overall structure of the paper, wrote the paper and approved the final version of the paper. F.B. reviewed the literature, contributed to figures and coded the experiments.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence should be addressed to Francisco Barreras or Duncan J. Watts.

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